relations

$$\begin{aligned}
&(v_1 + v_2) \otimes w - v_1 \otimes w - v_2 \otimes w, \\
&v \otimes (w_1 + w_2) - v \otimes w_1 - v \otimes w_2, \\
&va \otimes w - v \otimes aw, \quad a \in A.
\end{aligned}$$

Exercise 2.11.5. Let K be a field, and let L be an extension of K. If A is an algebra over K, show that $A \otimes_K L$ is naturally an algebra over L. Show that if V is an A-module, then $V \otimes_K L$ has a natural structure of a module over the algebra $A \otimes_K L$.

Problem 2.11.6. Throughout this problem, we let k be an arbitrary field (not necessarily of characteristic zero and not necessarily algebraically closed).

If A and B are two k-algebras, then an (A, B)**-bimodule** will mean a k-vector space V with both a left A-module structure and a right B-module structure which satisfy $(av)\, b = a\,(vb)$ for any $v \in V$, $a \in A$, and $b \in B$. Note that both the notions of "left A-module" and "right A-module" are particular cases of the notion of bimodules; namely, a left A-module is the same as an (A, k)-bimodule, and a right A-module is the same as a (k, A)-bimodule.

Let B be a k-algebra, W a left B-module, and V a right B-module. We denote by $V \otimes_B W$ the k-vector space $(V \otimes_k W) \,/\, \langle vb \otimes w - v \otimes bw \mid v \in V,\ w \in W,\ b \in B \rangle$. We denote the projection of a pure tensor $v \otimes w$ (with $v \in V$ and $w \in W$) onto the space $V \otimes_B W$ by $v \otimes_B w$. (Note that this tensor product $V \otimes_B W$ is the one defined in Remark 2.11.4.)

If, additionally, A is another k-algebra and if the right B-module structure on V is part of an (A, B)-bimodule structure, then $V \otimes_B W$ becomes a left A-module by $a\,(v \otimes_B w) = av \otimes_B w$ for any $a \in A$, $v \in V$, and $w \in W$.

Similarly, if C is another k-algebra, and if the left B-module structure on W is part of a (B, C)-bimodule structure, then $V \otimes_B W$ becomes a right C-module by $(v \otimes_B w)\, c = v \otimes_B wc$ for any $c \in C$, $v \in V$, and $w \in W$.